

The Unit Circle and Exact Values

ANSWER KEY



Exact values

1 Write the exact value of each expression:

a $\sin \frac{\pi}{6} = \frac{1}{2}$

b $\cos \frac{3\pi}{4} = -\frac{\sqrt{2}}{2}$

c $\tan \frac{5\pi}{6} = -\frac{\sqrt{3}}{3}$

d $\sin \frac{3\pi}{2} = -1$

2 Write the exact value of each expression:

a $\cos \left(-\frac{\pi}{3}\right) = \frac{1}{2}$

b $\sin \frac{7\pi}{4} = -\frac{\sqrt{2}}{2}$

c $\tan \pi = 0$

d $\cos \frac{11\pi}{6} = \frac{\sqrt{3}}{2}$

Points on the unit circle

3 The terminal side of angle θ in standard position passes through the point $P\left(-\frac{3}{5}, \frac{4}{5}\right)$ on the unit circle. Find:

a $\sin \theta = \frac{4}{5}$

b $\cos \theta = -\frac{3}{5}$

c $\tan \theta = -\frac{4}{3}$

d the quadrant containing θ Quadrant II

4 State the reference angle for each angle:

a $\frac{5\pi}{6} = \frac{\pi}{6}$

b $\frac{4\pi}{3} = \frac{\pi}{3}$

c $-\frac{\pi}{4} = \frac{\pi}{4}$

d $\frac{7\pi}{2} = \frac{\pi}{2}$

5 Identify the quadrant(s) in which each pair of conditions holds:

a $\sin \theta < 0$ and $\cos \theta > 0$ Quadrant IV.

b $\tan \theta > 0$ and $\sin \theta < 0$ Quadrant III.